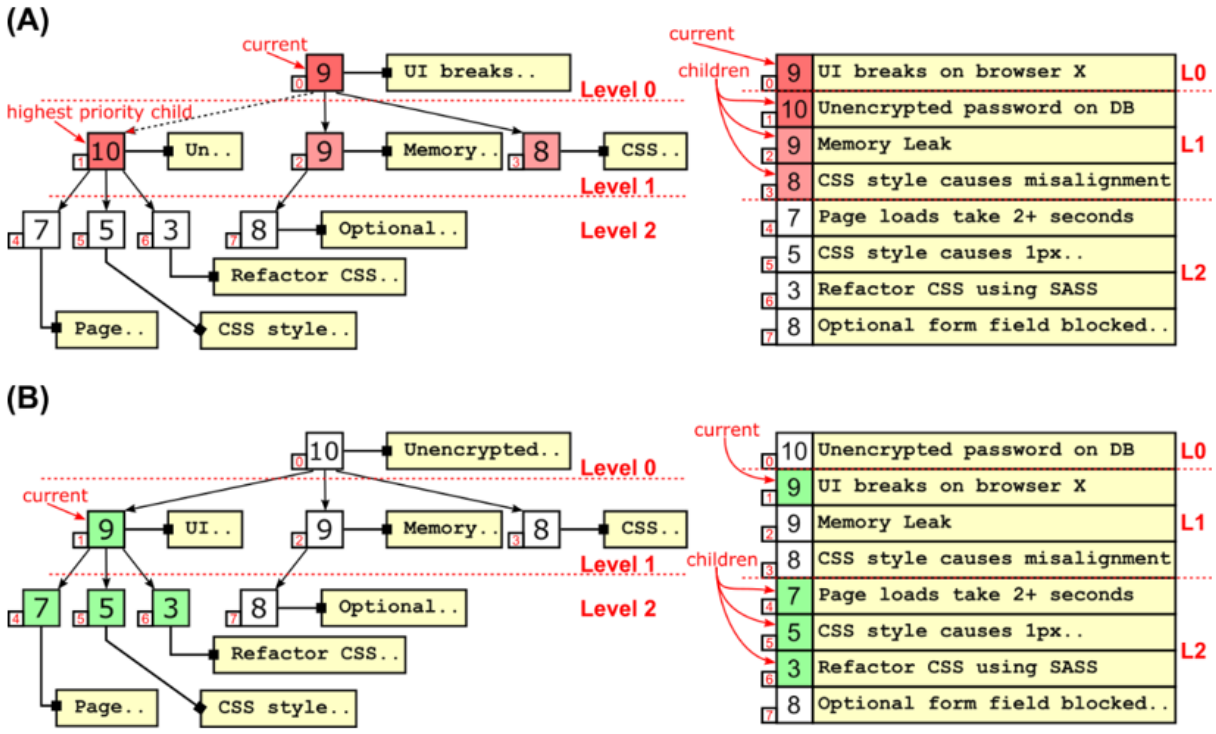


Figure 2.8 An example of applying the `pushDown` method to the root of a ternary heap.
(A) The root of the tree violates heap's properties, being larger than one of its children. The algorithm finds the child with the highest priority, and then compares it to current node (the root, in this example).
(B) We swapped the root with its highest-priority children: the node at index 1. Now, in turn, we compare the updated child with its children: since none of them has a higher priority than current node, we can stop.

To stick to our tasks example, let's consider the case where the root's priority is smaller than its children, shown in figure 2.8; working on a ternary Heap, its first leaf is stored at index 4.